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Surrogate Formulation for Chance-Constrained DC Optimal Power Flow With Affine Control Policy

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Abstract—In chance-constrained DC optimal power flow (CC-DCOPF) with affine control policy, the distribution of uncertainty variables is a function of the decision variable. Existing methods fail to transform chance constraints into an analytical form, resulting in computational difficulties. This letter proposes a surrogate formulation for CC-DCOPF considering the affine control policy. The data-driven surrogate model is constructed to identify the mapping relationship between the affine control variables and the tightening boundary of chance constraints. This allows us to analytically reformulate the chance constraints in the presence of non-Gaussian correlated uncertainties. By leveraging the physical power flow features, the surrogate model is scaled into a simple single function. This enables us to use the piecewise linearization technique to approximate the CC-DCOPF to a mixed-integer linear programming (MILP) problem. Case studies show that the proposed method leads to a near-optimal solution with high computational efficiency.

Index Terms—DC optimal power flow, chance constraints, surrogate model, renewable energy, data-driven.

I. INTRODUCTION

AS A representative stochastic optimization method, chance-constrained optimization has been widely used for uncertainty management in DC optimal power flow (DCOPF) [1]. For the chance-constrained DC optimal power flow (CC-DCOPF) problem, existing methods can be generally classified into sampling-based methods and distribution-based methods. In the sampling-based methods, the probabilities of chance constraints are estimated by sampling techniques [2], [3]. Because the optimality is ensured with the detailed modeling of each scenario, the computational burden quickly rises with the number of scenarios. In the distribution-based methods, the CC-DCOPF is usually formulated as a deterministic optimization problem based on specific probabilistic distribution assumptions [4]. Data-driven nonparametric uncertainty modeling methods, such

as the *Gaussian mixture model* (GMM) [5], can be applied to address the distribution assumption issues. This is achieved by modeling the tightening boundary of chance constraints as the inverse *cumulative distribution functions* (CDFs) of uncertainty variables with a pre-determined affine control policy, which is estimated from a database. An iterative approach is proposed in [6] to find a suitable security parameter to guarantee the feasibility of the CC-DCOPF problem under a non-Gaussian uncertainty distribution.

The affine control policy modeled by the automatic generation control (AGC) participation factors can be used to reflect the adjustment of conventional generators for compensating the power mismatch caused by uncertainties. However, it brings a formulation challenge in the CC-DCOPF modeling because the uncertainty variables, such as transmission power flows and generation outputs, will be influenced by the AGC participation factors. It is difficult to analytically express the inverse CDFs in the form of the tightening boundary of chance constraints because the CDFs of uncertainty variables become a function of the AGC participation factors. To the best of our knowledge, there is still no solution yet for the analytical expression in the presence of non-Gaussian uncertainty variables considering the AGC participation factors. In this letter, a data-driven method for the CC-DCOPF reformulation with non-Gaussian uncertainties is developed. Instead of modeling the CDFs, a data-driven surrogate model is constructed to directly identify the tightening boundary, which can be used as a function of AGC participation factors in the CC-DCOPF. Furthermore, the data-driven surrogate model is scaled into a simple single-function based on the physical power flow features, avoiding the high-dimensional mapping relationship which is established in the learning process from the AGC participation factors to the tightening boundary of chance constraints. This enables us not only to build an accurate surrogate model but also to use the piecewise linearization technique to guarantee the convexity of the CC-DCOPF problem.

II. PROBLEM FORMULATION

The classic CC-DCOPF model is formulated as follows (the single chance-constraint formulation is used here) [5]:

$$f = \min_{P_{sc,i}, \beta_i} E \left(\sum_{i \in I_G} c_i P_{sc,i} \right), \quad (1a)$$

$$\text{s.t.} \quad \sum_{i \in I_G} P_{sc,i} - \sum_{i \in I_{PI}} (D_{fc,i} - W_{fc,i}) = 0, \quad (1b)$$

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$$\Pr\left(\sum_{i \in I_G} PTDF_{li}P_i - \sum_{i \in I_{PI}} PTDF_{li}(D_i - W_i) \leq F_l\right) \geq 1 - \varepsilon, \quad (1c)$$

$$\Pr\left(-\sum_{i \in I_G} PTDF_{li}P_i + \sum_{i \in I_{PI}} PTDF_{li}(D_i - W_i) \leq F_l\right) \geq 1 - \varepsilon, \quad (1d)$$

$$\Pr(P_i \leq P_{\max,i}) \geq 1 - \varepsilon, \quad (1e)$$

$$\Pr(-P_i \leq -P_{\min,i}) \geq 1 - \varepsilon, \quad (1f)$$

$$P_i = P_{sc,i} + \Delta P_i, D_i = D_{fc,i} + \Delta D_i, W_i = W_{fc,i} + \Delta W_i, \quad (1g)$$

$$\Delta P_i = \beta_i \sum_{i \in I_{PI}} (\Delta D_i - \Delta W_i), \quad (1h)$$

$$0 \leq \beta_i \leq 1, \sum_{i \in I_G} \beta_i = 1. \quad (1i)$$

where c_i is the generator cost; P_i , $P_{sc,i}$, ΔP_i are actual generation output, scheduled generation output, and generation response to balance the system uncertainty, respectively; D_i , $D_{fc,i}$, ΔD_i and W_i , $W_{fc,i}$, ΔW_i are actual value, forecasted value, and the forecasted errors of demand and wind power, respectively; i and l are the indices of bus and transmission lines; β_i is the AGC participation factor; $PTDF_{li}$ represents the power transfer distribution factor of bus i to line l ; ε is the pre-defined confidence level; I_{PI} and I_G are the sets of power injections and thermal generators, respectively.

In general, the chance constraints (1c)–(1f) can be reformulated as

$$\sum_{i \in I_G} PTDF_{li}P_{sc,i} - \sum_{i \in I_{PI}} PTDF_{li}(D_{fc,i} - W_{fc,i}) + UM_{up,l}^{PF}(\beta) \leq F_l, \quad (2a)$$

$$-\sum_{i \in I_G} PTDF_{li}P_{sc,i} + \sum_{i \in I_{PI}} PTDF_{li}(D_{fc,i} - W_{fc,i}) + UM_{dn,l}^{PF}(\beta) \leq F_l, \quad (2b)$$

$$P_i + UM_{up,i}^G(\beta_i) \leq P_{\max,i}, \quad (2c)$$

$$-P_i + UM_{dn,i}^G(\beta_i) \leq -P_{\min,i}, \quad (2d)$$

where $UM(\beta)$ represents the tightening boundary of the specific chance constraint, namely *uncertainty margin* (UM), which is correlated with β and system uncertainties. Usually, the chance constraints (2a)–(2d) are hard to be analytically reformulated in the presence of non-Gaussian correlated uncertainties due to the lack of closed-form of the UM. In existing methods, with a pre-determined β , an approximate UM is established by using the nonparametric modeling method to create the inverse CDFs of uncertainty variables [5]. Thus, the CC-DCOPF can be transformed into a linear programming problem. However, β is a dispatch variable and keeps changing in the optimization process. Therefore, it is extremely difficult, not to say impossible, to build the inverse CDFs under the condition of countiously changing AGC participation factors.

To address this challenging issue, a data-driven surrogate model-based CC-DCOPF method is proposed. From a data-driven perspective, the UM can be regarded as a mapping function of β , which can be approximated using historical data.

Hence, the key idea is to construct a data-driven surrogate model $f: \beta \rightarrow UM(\beta)$ to obtain the analytical closed-form of $UM(\beta)$. Then, by embedding the surrogated UM into (2a)–(2d), the CC-DCOPF with the affine control policy can be solved by an off-the-shelf solver. It is worth noting that some other methods, such as the robust optimization, can be also used to handle various stochastic optimization problems [7]. Different methods have distinct features and advantages/disadvantages. We only focus on the chance-constrained approach in this letter.

III. PROPOSED SURROGATE MODEL-BASED CC-DCOPF

A. Basic Idea

In the CC-DCOPF model in (1), the transmission power flows and generator outputs are the uncertainty variables, which can be formulated by considering (1g)–(1h) as follows:

$$PF_l = \sum_{i \in I_G} PTDF_{li}P_{sc,i} - \sum_{i \in I_{PI}} PTDF_{li}D_{fc,i} + \sum_{i \in I_{PI}} \sum_{k \in I_G} (PTDF_{lk}\beta_k - PTDF_{li})(\Delta D_i - \Delta W_i) \quad (3)$$

$$P_i = P_{sc,i} + \beta_i \sum_{i \in I_{PI}} (\Delta D_i - \Delta W_i) \quad (4)$$

The uncertainty part of transmission power flow ΔPF_l is defined as:

$$\Delta PF_l = \sum_{i \in I_{PI}} \sum_{k \in I_G} (PTDF_{lk}\beta_k - PTDF_{li})(\Delta D_i - \Delta W_i) \quad (5)$$

Similarly, ΔP_i in (1h) represents the uncertainty part of generator output. The UM can be formulated as a function of β :

$$UM_{up,l}^{PF}(\beta) = \Gamma^{1-\varepsilon}(\Delta PF_l) \quad (6a)$$

$$UM_{dn,l}^{PF}(\beta) = \Gamma^{1-\varepsilon}(-\Delta PF_l) \quad (6b)$$

$$UM_{up,i}^G(\beta_i) = \Gamma^{1-\varepsilon}(\Delta P_i) \quad (6c)$$

$$UM_{dn,i}^G(\beta_i) = \Gamma^{1-\varepsilon}(-\Delta P_i) \quad (6d)$$

where $\Gamma^{1-\varepsilon}(\cdot)$ represents the $(1-\varepsilon)$ -quantile function of the uncertainty variables. Note that although the expression of UM is not analytical due to the quantile function $\Gamma^{1-\varepsilon}(\cdot)$, generation training samples can be used to build the surrogate model by a simple Monte Carlo simulation (MCS) process.

Taking $UM_{up,l}^{PF}(\beta)$ as an example, it can be formulated as:

$$UM_{up,l}^{PF}(\beta) = \Gamma^{1-\varepsilon}\left(\sum_{i \in I_{PI}} \sum_{k \in I_G} (PTDF_{lk}\beta_k - PTDF_{li})(\Delta D_i - \Delta W_i)\right) \quad (7)$$

The basic motivation of this letter is to construct surrogates for β and $UM_{up,l}^{PF}(\beta)$ to obtain the analytical form of $UM_{up,l}^{PF}(\beta)$. The technical challenges for the data-driven surrogate modeling of $UM_{up,l}^{PF}(\beta)$ are elaborated as follows. First, to train an accurate surrogate model, it is essential to generate enough

¹ $\Gamma^{1-\varepsilon}(\Delta PF_l)$ is defined as the upper quantile corresponding to the $(\varepsilon N)^{\text{th}}$ largest magnitude $\Delta PF_l^{\varepsilon N}$. Similarly, $\Gamma^{1-\varepsilon}(-\Delta PF_l)$ is defined as the lower quantile corresponding to the $(1-\varepsilon N)^{\text{th}}$ lowest magnitude $\Delta PF_l^{(1-\varepsilon)N}$.

samples to uniformly fulfill the feasible region of β in (1i). This is a challenging task for large-scale systems. Second, the expression of the surrogate of $UM_{up,l}^{PF}(\beta)$ is a multi-variable non-linear function. The linearization of the analytical model obtained from the data-driven framework is not an easy task either.

For each transmission line, the sensitivities of each node's uncertainties $(\Delta D_i - \Delta W_i)$ are the same, which is $\sum_{k \in I_G} PTDF_{lk} \beta_k$. Based on this fact, the surrogate function $UM_{up,l}^{PF}(\beta)$ can be scaled into a one-variable function so that the above-mentioned challenges can be overcome. This is because it is significantly simplified to generate the training samples in a one-dimension feasible region and to use the accurate piecewise linearization technique to maintain the linearity of CC-DCOPF in a one-dimension space.

B. Surrogate Model-Based Chance Constraint Formulation

For the UM of the transmission power flow, reformulating (7) yields

$$UM_{up,l}^{PF}(\kappa_l) = \Gamma^{1-\varepsilon} \left(\sum_{i \in I_{PI}} (\kappa_l - PTDF_{li})(\Delta D_i - \Delta W_i) \right) \quad (8)$$

$$\kappa_l = \sum_{k \in I_G} PTDF_{lk} \beta_k \quad (9)$$

A new one-dimension variable κ_l is introduced, which represents the sensitive factors of the variation of transmission power flow caused by random generator output with respect to ΔD_i . The feasible region (i.e., the upper and lower limits) of κ_l can be obtained by solving the following simple LP problem for surrogate model:

$$\kappa_{l,\max} = \max_{\beta_i} (\sum_{k \in I_G} PTDF_{lk} \beta_k) \quad (10)$$

s.t. constraints (1i).

$$\kappa_{l,\min} = \min_{\beta_i} (\sum_{k \in I_G} PTDF_{lk} \beta_k) \quad (11)$$

s.t. constraints (1i).

As seen in (8), the UM of transmission power flow constraint is formulated as a one-variable function of κ_l , making it much easier to construct the surrogate model. By taking κ_l as the input and the UM of transmission power flow as the output, the Kriging surrogate method is utilized to approximate the closed-form of the UM function with the linear trend and exponential correlation [8], as expressed below:

$$UM_{Kriging,up,l}^{PF}(\kappa_l) = \chi^T(\kappa_l) \alpha_{up,l} + \mathbf{r}^T(\kappa_l) \mathbf{C}^{-1}(\mathbf{y}_{t,up,l} - \Psi_{up,l} \alpha_{up,l}) \quad (12)$$

where χ and Ψ are the trend function vector and matrix, respectively; α is the corresponding trend coefficients; $\mathbf{r}$ and $\mathbf{C}$ are the correlation function vector and matrix, respectively. Note that the proposed method does not rely on a certain type of learning tool. Kriging surrogate is used because of its straightforward structure for analytical modeling.

The surrogate UMs are nonlinear functions. Fortunately, $UM_{Kriging,up,l}^{PF}(\kappa_l)$ is an one-variable function of κ_l . By partitioning the feasible range of variable κ_l into several uniform segments, $UM_{Kriging,up,l}^{PF}(\kappa_l)$ can be piecewise linearized to maintain the convex form of the problem. The piecewise linear UMs are denoted as $UM_{pl,up,l}^{PF}(\kappa_l)$, where subscript pl is an integer index representing the segment No. in the piecewise

linear curve. In a similar way, the linearized form of $UM_{dn,l}^{PF}(\beta)$ can be denoted as $UM_{pl,dn,l}^{PF}(\kappa_l)$.

For the generator output constraints, by substituting (1h) into (6c) and (6d), the $UM_{up,l}^G(\beta_i)$ and $UM_{dn,l}^G(\beta_i)$ are formulated as:

$$UM_{up,i}^G(\beta_i) = \beta_i \Gamma^{1-\varepsilon} \left(\sum_{i \in I_{PI}} (\Delta D_i - \Delta W_i) \right) \quad (13)$$

$$UM_{dn,i}^G(\beta_i) = \beta_i \Gamma^{1-\varepsilon} \left(\sum_{i \in I_{PI}} (\Delta W_i - \Delta D_i) \right) \quad (14)$$

Note that the terms $\Gamma^{1-\varepsilon} (\sum_{i \in I_{PI}} (\Delta D_i - \Delta W_i))$ and $\Gamma^{1-\varepsilon} (\sum_{i \in I_{PI}} (\Delta W_i - \Delta D_i))$ remain unchanged, which can be directly obtained by MCS in advance. The convexity is naturally guaranteed. After implementing the proposed chance constraint formulation method the new CC-DCOPF model is formulated in the following, by substituting $UM_{pl,up,l}^{PF}(\kappa_l)$ and $UM_{pl,dn,l}^{PF}(\kappa_l)$ for $UM_{up,l}^{PF}(\beta)$ and $UM_{dn,l}^{PF}(\beta)$ in Eqs. (2a) and (2b), respectively:

$$f = \min_{P_{sc,i}, \beta_i} E \left(\sum_{i \in I_G} c_i P \right) \quad (15)$$

s.t., constraints (1b), (1i), (2a)–(2d).

The terms $UM_{pl,up,l}^{PF}(\kappa_l)$ and $UM_{pl,dn,l}^{PF}(\kappa_l)$ make the proposed model become a *mixed-integer linear programming* (MILP) problem, which can be easily solved without convergence issues. To reduce the number of integer variables in transmission constraints, a successive constraint identification method is applied to solve the MILP. The successive constraint identification method is summarized as follows. First, the CC-DCOPF in the MILP form, including only the active transmission constraints in the deterministic DC-OPF is solved. Second, whether the obtained solution satisfies all transmission constraints is checked. Third, the violated transmission constraints are added back to the problem, and the MILP is resolved until the convergence criterion is reached.

IV. NUMERICAL RESULTS

In this section, the proposed method is tested on the IEEE 118-bus system with 5 wind farms. The realistic wind power data is obtained from Bonneville Power Administration [9]. The loads are assumed to follow a Gaussian distribution with standard deviations equaling 5% of the original loads. Three hundred training samples are generated for the surrogate modeling. The simulations are performed on a PC equipped with Intel Core i7-8700 CPU @ 3.20 GHz 32 GB RAM.

The following methods are compared.

- M0: the proposed method;
- M1: the sampling-based method proposed in [2];
- M2: the improved sampling-based method according to [3]²;
- M3: GMM-based method [5] (the AGC participation factors are fixed according to the maximum outputs of generators).

A. Performance Under Various Scenarios

To demonstrate the effectiveness of the proposed method, M0 is compared with two scenarios-based methods (M1 and M2)

²M2 is implemented by reducing the scale of the Big-M model to the same level as in Ref. [3].

TABLE I
EFFECTIVENESS EVALUATION OF DIFFERENT METHODS

	Number of samples	100	500	1000	10000
M0	Generation cost (10^4 \$)	8.8865	8.1929	9.8059	9.8992
	MIP-Gap (%)	0	0	0	0
	Calculation time (s)	4.32	4.46	4.23	4.34
M1	Generation cost (10^4 \$)	8.8841	-	-	-
	MIP-Gap (%)	0.7305	-	-	-
	Calculation time (s)	10000*	-	-	-
M2	Generation cost (10^4 \$)	8.8832	8.1928	9.8055	-
	MIP-Gap (%)	0	0	0	-
	Calculation time (s)	12.39	20.92	29.82	-

"-": MIP problem runs out of memory; "*": case hits the 10000s time limit.

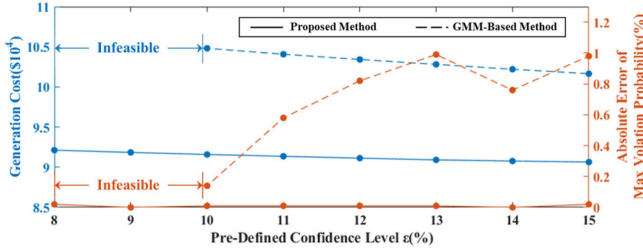


Fig. 1. Performance comparison under different confidence levels.

under the different numbers of scenarios, as shown in Table I. The pre-defined confidence level ε and the computational time limit are set to be 0.05 and 10000s

Several conclusions can be drawn as follows:

- 1) Compared with the scenarios-based methods, the proposed method can achieve a near-optimal solution. With the increasing number of scenarios, M1 and M2 will need to increase the number of involved integer variables. By contrast, the proposed method can conveniently handle this situation because the problem scale of the proposed method (influenced by the *piecewise linear segments* $\times$ *the number of transmission constraints*) does not rely on the number of scenarios. Meanwhile, the involved integer variables can be further reduced in each step of the iterative solution process using the proposed successive constraint identification method.
- 2) By comparing M1 and M2, we can observe that for the sampling-based method, the calculation performance can be greatly improved using proper techniques, like the constraint screening and the tighten of Big-M value in [3]. The proposed method presents a different approach to address this CC-OPF problem using the assistance of data-driven tools. The implementation is straightforward, providing a new perspective to solve this problem.

B. Performance With Different Confidence Levels

The proposed method is tested with the confidence levels ranging from 8% to 15%. The number of uncertainty scenarios used for both the CC-DCOPF modeling and the max constraint violation probabilities testing is set to 10000. The results are shown in Fig. 1. The sampling-based methods presented by M1 and M2 can achieve the optimality given sufficient computational time, which can be regarded as the benchmark. Here, the proposed method is compared with the distribution-based method (M3), as shown in Fig. 1.

The blue line represents the generation cost under different confidence levels. We can see that the generation costs gradually

decrease as the pre-defined confidence level increases, which means that lower confidence achieves more conservative solution results. As can be seen from Fig. 1, the GMM-based method cannot solve the cases with confidence levels ranging under 10% because the AGC participation factors cannot be optimized by the GMM-based method. Compared with the GMM-based method (dashed lines in Fig. 1), the proposed method (solid lines in Fig. 1) can find the solution under a stricter confidence level while having a lower generation cost. This is because the AGC participation factors optimization provides extra flexibility to handle uncertainties.

The orange line represents the absolute error of max violation probability. For the GMM-based method, the absolute error of max violation probability is relatively large, which comes from the GMM-based modeling for uncertainties distribution. The GMM-based method approximates the uncertainty distribution based on the historical data to obtain the CDFs of uncertainty variables for CC-DCOPF modeling. The approximate error leads to the max violation probability diverting from the pre-defined confidence level. By comparison, the proposed method (solid lines in Fig. 1) directly constructs the surrogate model of uncertainty margin by historical data. The uncertainty modeling error is avoided, achieving a lower absolute error of max violation probability than the GMM-based methods. It shows better feasibility for the required confidence level.

V. CONCLUSION

This letter proposes a data-driven surrogate model based method for the CC-DCOPF problem considering the affine control policy. To the best of our knowledge, the proposed method is the first CC-DCOPF method considering the continuous changes of AGC participation factors in the optimization process without any uncertainty distribution assumptions. The performance is justified in the case studies. The proposed method provides extra flexibility to handle uncertainties in the CC-DCOPF problem.

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